

state network connections, based on the most secure Wi-Fi security type available within beacons and probe responses for the set of all discovered BSSIDs for the configured SSID. The type of PSK or passphrase used SHALL be inferred based on the length and contents of the Credentials field provided, matching the security type chosen.

Valid Credentials length are:

- 0 bytes: Unsecured (open) connection
- 5 bytes: WEP-64 passphrase
- 10 hexadecimal ASCII characters: WEP-64 40-bit hex raw PSK
- 13 bytes: WEP-128 passphrase
- 26 hexadecimal ASCII characters: WEP-128 104-bit hex raw PSK
- 8..63 bytes: WPA/WPA2/WPA3 passphrase
- 64 bytes: WPA/WPA2/WPA3 raw hex PSK

These lengths SHALL be contextually interpreted based on the security type of the BSSID where connection will occur.

When the length of Credentials and available set of BSSID admits more than one option, such as the presence of both WPA2 and WPA security type within the result set, WPA2 SHALL be considered more secure.

Note that it MAY occur that a station cannot connect to a particular access point with higher security and selects a lower security connectivity type if the link quality is deemed to be too low to achieve successful operation, or if all retry attempts fail.

Breadcrumb Field

See [Section 11.9.7.1.2, “Breadcrumb Field”](#) for usage.

11.9.7.4. AddOrUpdateThreadNetwork Command

This command is used to add or update a Thread network configuration.

This command SHALL be used to add or modify Thread network configurations.

If this command is received without an armed fail-safe context (see [Section 11.10.7.2, “ArmFail-Safe”](#)), then this command SHALL fail with a FAILSAFE_REQUIRED status code sent back to the initiator.

See [Section 11.9.7.5, “Common processing of AddOrUpdateWiFiNetwork and AddOrUpdateThreadNetwork”](#) for behavior of addition/update.